

HDOC DAY : 32

Aim: To write a menu-driven C program to demonstrate that matrix addition is commutative ($A+B=B+A$) and matrix multiplication is non-commutative ($A * B \neq B * A$) using functions and variable-length 2D arrays.

Algorithm:

1. Start

2. Define Functions:

- a. readMatrix: Reads rows, columns, and elements of a 2D array from the user.
- b. printMatrix: Prints the elements of a 2D array in matrix format.
- c. addMatrices: Adds two matrices and stores the result in a third matrix.
- d. multiplyMatrices: Multiplies two matrices and stores the result in a third matrix.
- e. areEqual: Checks if two matrices have identical elements.

3. Menu Loop (do-while):

- a. Display options:
 - i. Prove Matrix Addition is Commutative
 - ii. Prove Matrix Multiplication is Not Commutative
 - iii. Exit
- b. Input user choice.

4. If Choice = 1:

- a. Input dimensions for Matrix A and Matrix B.
- b. If dimensions do not match ($r1 \neq r2$ or $c1 \neq c2$), display an error message.
- c. Otherwise, input elements for Matrix A and Matrix B.
- d. Compute $Sum1=A+B$ and $Sum2=B+A$.

- e. Compare Sum1 and Sum2. If they are equal, state that matrix addition is commutative.

5. If Choice = 2:

- a. Input dimensions for Matrix A ($r1 \times c1$) and Matrix B ($r2 \times c2$).
- b. Check if multiplication condition holds ($c1=r2$ and $c2=r1$). If not, display an error message.
- c. Input elements for Matrix A and Matrix B.
- d. Compute $Prod1=A \times B$ and $Prod2=B \times A$.
- e. Compare Prod1 and Prod2. If they are not equal, state that matrix multiplication is not commutative.

6. If Choice = 3:

- a. Exit the program.

7. Stop

C code :

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File: 32.c

#include <stdio.h>

void inputMatrix(int r, int c, int mat[r][c]) {
    for (int i = 0; i < r; i++) {
        for (int j = 0; j < c; j++) {
            scanf("%d", &mat[i][j]);
        }
    }
}

void displayMatrix(int r, int c, int mat[r][c]) {
    for (int i = 0; i < r; i++) {
        for (int j = 0; j < c; j++) {
            printf("%d ", mat[i][j]);
        }
        printf("\n");
    }
}

void add(int r, int c, int A[r][c], int B[r][c], int ans[r][c]) {
    for (int i = 0; i < r; i++) {
        for (int j = 0; j < c; j++) {
            ans[i][j] = A[i][j] + B[i][j];
        }
    }
}

void multiply(int r1, int c1, int A[r1][c1], int r2, int c2, int B[r2][c2], int ans[r1][c2]) {
    for (int i = 0; i < r1; i++) {
        for (int j = 0; j < c2; j++) {
            ans[i][j] = 0;
            for (int k = 0; k < c1; k++) {
                ans[i][j] += A[i][k] * B[k][j];
            }
        }
    }
}

int checkEqual(int r, int c, int A[r][c], int B[r][c]) {
    for (int i = 0; i < r; i++) {
        for (int j = 0; j < c; j++) {
            if (A[i][j] != B[i][j]) {
                return 0;
            }
        }
    }
    return 1;
}

int main() {
    int choice;

    do {
        printf("\n1. Matrix Addition Commutative Test\n");
        printf("\n2. Matrix Multiplication Commutative Test\n");
    } while (choice != 0);
}
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do {
    printf("\n1. Matrix Addition Commutative Test\n");
    printf("2. Matrix Multiplication Commutative Test\n");
    printf("3. Exit\n");
    printf("Enter choice: ");
    scanf("%d", &choice);

    if (choice == 1) {
        int r1, c1, r2, c2;
        printf("Enter rows and cols for Matrix A: ");
        scanf("%d %d", &r1, &c1);
        printf("Enter rows and cols for Matrix B: ");
        scanf("%d %d", &r2, &c2);

        if (r1 != r2 || c1 != c2) {
            printf("Addition not possible! Dimensions must be same.\n");
        } else {
            int A[r1][c1], B[r2][c2];
            int sum1[r1][c1], sum2[r1][c1];

            printf("Enter elements of Matrix A:\n");
            inputMatrix(r1, c1, A);

            printf("Enter elements of Matrix B:\n");
            inputMatrix(r2, c2, B);

            add(r1, c1, A, B, sum1);
            add(r1, c1, B, A, sum2);

            printf("\nMatrix A + B:\n");
            displayMatrix(r1, c1, sum1);

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            printf("Matrix B + A:\n");
            displayMatrix(r1, c1, sum2);

            if (checkEqual(r1, c1, sum1, sum2) == 1) {
                printf("Result: Matrix Addition is Commutative (A + B = B + A)\n");
            } else {
                printf("Result: Not Commutative\n");
            }
        }
    }
    else if (choice == 2) {
        int r1, c1, r2, c2;
        printf("Enter rows and cols for Matrix A: ");
        scanf("%d %d", &r1, &c1);
        printf("Enter rows and cols for Matrix B: ");
        scanf("%d %d", &r2, &c2);

        if (c1 != r2 || c2 != r1) {
            printf("Both A*B and B*A are not possible with these sizes.\n");
        } else {
            int A[r1][c1], B[r2][c2];
            int prod1[r1][c2], prod2[r2][c1];

            printf("Enter elements of Matrix A:\n");
            inputMatrix(r1, c1, A);

            printf("Enter elements of Matrix B:\n");

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        printf("Enter elements of Matrix B:\n");
        inputMatrix(r2, c2, B);

        multiply(r1, c1, A, r2, c2, B, prod1);
        multiply(r2, c2, B, r1, c1, A, prod2);

        printf("\nMatrix A * B:\n");
        displayMatrix(r1, c2, prod1);

        printf("Matrix B * A:\n");
        displayMatrix(r2, c1, prod2);

        if (checkEqual(r1, c2, prod1, prod2) == 0) {
            printf("Result: Matrix Multiplication is NOT Commutative ( $A * B \neq B * A$ )\n");
        } else {
            printf("Result: Matrices are equal.\n");
        }
    }
}
else if (choice == 3) {
    printf("Exiting program...\n");
}
else {
    printf("Invalid choice!\n");
}
} while (choice != 3);

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    return 0;
}

```

SCREENSHOT :

```
[manalithakar@Unknown_0a:57:f1:e7:bf:b0 ~ % gcc 32.c -o 32  
[manalithakar@Unknown_0a:57:f1:e7:bf:b0 ~ % ./32
```

1. Matrix Addition Commutative Test
2. Matrix Multiplication Commutative Test
3. Exit

Enter choice: 1

Enter rows and cols for Matrix A: 2 2

Enter rows and cols for Matrix B: 2 2

Enter elements of Matrix A:

1 2

3 4

Enter elements of Matrix B:

5 6

7 8

Matrix A + B:

6 8

10 12

Matrix B + A:

6 8

10 12

Result: Matrix Addition is Commutative ($A + B = B + A$)

1. Matrix Addition Commutative Test
2. Matrix Multiplication Commutative Test
3. Exit

Enter choice: 2

Enter rows and cols for Matrix A: 2 2

Enter rows and cols for Matrix B: 2 2

Enter elements of Matrix A:

1 2

3 4

Enter elements of Matrix B:

5 6

7 8

Matrix A * B:

19 22

43 50

Matrix B * A:

23 34

31 46

Result: Matrix Multiplication is NOT Commutative ($A * B \neq B * A$)

1. Matrix Addition Commutative Test
2. Matrix Multiplication Commutative Test
3. Exit

Enter choice: 3

Exiting program...

```
manalithakar@Unknown_0a:57:f1:e7:bf:b0 ~ % █
```